Home Service Booking Platform Based on Service-Oriented Architecture (SOA)

1. Introduction

Finding trustworthy service providers such as electricians, plumbers, cleaners, painters, and technicians is often challenging. Customers usually rely on personal recommendations, social media posts, or local advertisements, which may not always be reliable.

The proposed **Home Service Booking Platform** aims to provide a centralized online platform where customers can search for service providers, book services, track booking status, and receive notifications. Service providers can manage their profiles, availability, and service requests.

The system will be developed using **Service-Oriented Architecture (SOA)** principles, where independent services communicate through REST APIs while maintaining loose coupling.

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2. Problem Statement

Current home service booking processes suffer from several issues:

- Difficulty finding trusted service providers
- Lack of centralized booking management
- Poor communication between customers and providers
- No booking tracking mechanism
- Manual scheduling conflicts
- Limited service history and accountability

A Service-Oriented Architecture can address these issues by separating business functionalities into independent services.

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3. Project Objectives

Main Objective

To develop a service-oriented platform that efficiently connects customers with service providers while ensuring seamless booking management and communication.

Specific Objectives

- Allow service providers to register and manage profiles.
- Allow customers to search for available services.
- Enable customers to book services online.
- Track booking progress and status.
- Send notifications regarding booking updates.
- Integrate with an external payment system.
- Maintain service history for customers and providers.

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4. Scope of the System

Service Categories

- Electrician
- Plumber
- Cleaner
- Carpenter
- Painter
- AC Technician
- CCTV Technician
- Appliance Repair

Customer Functions

- Register/Login
- Search Service Providers
- Book Services
- View Booking History
- Track Booking Status
- Receive Notifications

Service Provider Functions

- Register/Login
- Manage Profile
- Set Availability
- Accept/Reject Bookings
- View Service History

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5. System Architecture

The proposed solution consists of **three independent microservices** and **one external conceptual system**.

Service 1: Provider Management Service

Responsibilities

- Register service providers
- Manage provider profiles
- Maintain service categories
- Manage provider availability
- Store provider ratings

Database

Field
ProviderID
Name
ContactNumber
Email
ServiceType
Location
Availability
Rating

APIs

```
POST /providers
GET /providers
GET /providers/{id}
PUT /providers/{id}
DELETE /providers/{id}
```

Service 2: Booking Management Service

Responsibilities

- Create service bookings
- Manage booking schedules
- Assign service providers
- Track booking progress
- Maintain booking history

Booking Status

- Pending
- Accepted
- In Progress
- Completed
- Cancelled

Database

Field
BookingID
CustomerID
ProviderID
ServiceType
BookingDate
Address
Status

APIs

```
POST /bookings
GET /bookings
GET /bookings/{id}
PUT /bookings/status
DELETE /bookings/{id}
```

Service 3: Notification Service

Responsibilities

- Generate booking notifications
- Send acceptance notifications

- Send rejection notifications
- Send completion notifications
- Send payment notifications

Notification Examples

- Booking Created
- Booking Accepted
- Booking Rejected
- Service Completed
- Payment Successful

Database

Field
NotificationID
UserID
Message
Status
Timestamp

APIs

```
POST /notifications
GET /notifications/{userId}
```

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6. External Conceptual System

Payment Gateway System

Purpose

The Home Service Booking Platform will conceptually interact with an external payment gateway.

Functions

- Process online payments
- Verify payment transactions
- Return payment status
- Generate payment confirmation

Benefits

- Secure transactions
- Reduced cash handling
- Faster payment processing

Example

Before confirming a booking, the system verifies payment through the payment gateway.

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7. Service Interactions

Scenario 1: Service Provider Registration

```
Provider
    |
    v
Provider Management Service
    |
Provider Database
```

Scenario 2: Customer Books a Service

```
Customer
    |
    v
Booking Management Service
    |
Booking Stored
    |
    v
Notification Service
    |
Booking Confirmation Sent
```

Scenario 3: Provider Accepts Booking

```
Provider
    |
    v
Booking Management Service
    |
Booking Status Updated
    |
```

```
    v
Notification Service
    |
Customer Notified
```

Scenario 4: Customer Makes Payment

```
Customer
    |
    v
Booking Management Service
    |
    v
Payment Gateway
    |
Payment Verified
    |
Booking Confirmed
```

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8. Technologies

Backend

- Java Spring Boot

Database

- MySQL

API Communication

- REST APIs
- JSON

Security

- JWT Authentication
- Password Hashing

Frontend

- React.js

or

- HTML/CSS/JavaScript

Service Testing

- Postman

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9. Expected Outcomes

The completed system will:

- Simplify the process of finding service providers.
- Improve booking management efficiency.
- Provide secure online payment integration.
- Enhance communication between customers and providers.
- Demonstrate Service-Oriented Architecture concepts.
- Showcase loose coupling and independent service deployment.

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10. Conclusion

The **Home Service Booking Platform** is a practical business-oriented application that demonstrates Service-Oriented Architecture through independent **Provider Management**, **Booking Management**, and **Notification Services**. Integration with an external **Payment Gateway System** further enhances realism and business value. The project effectively demonstrates SOA principles such as loose coupling, service independence, interoperability, scalability, and maintainability while remaining achievable within a university project timeline.